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AQEval: R code for the analysis of discrete change in Air Quality time-series

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Summary

AQEval (Air Quality Evaluation) is an R package for the routine investigation of discrete changes in air quality time-series. The main functions, quantBreakPoints and quantBreakSegments use a three-step method to find possible 'points-of-change', test these and quantify the most likely or 'regions-of-change' about them, respectively. Other key functions build of these to provide a workflow to measure smaller changes and/or changes in more complex environments.

Statement of Need

Authorities responsible for air quality management are typically required to implement and evaluate air quality interventions they adopt (Bradley et al., 2019). These interventions are often costly, disruptive and unpopular (Glazener & Khreis, 2019), and the inherent variability monitoring data hinders impact assessments (Grange & Carslaw, 2019; Jones et al., 2012; Kelly et al., 2011; Pearce et al., 2011). Various methods have been developed to investigate discrete changes in a wide range of time-series (see e.g. Reeves et al., 2007; Truong et al., 2020) and several R (R Core Team & others, 2025) packages have been developed for their use, e.g. bcp (Erdman & Emerson, 2008), changepoint (Killick et al., 2016), segmented (Muggeo & others, 2008), and strucchange (Zeileis et al., 2002). Some have even been applied to air quality time-series (see e.g. Carslaw et al. (2006), Carslaw & Carslaw (2007)). However, many of those tasked with air quality policy assessment, although highly skilled in a wide range of monitoring activities, are unlike to be able to dedicate sufficient time and resources to the development of in-house expertise in such specialist analyses. AQEval was developed to address this skill gap. It aligns the inputs and outputs of a number of statistical methods to provide a one-package option for anyone interested in using R to routinely investigate change in air quality data.

Sources

AQEval is freely available under General Public License (GPL):

- The latest (stable) release version of AQEval is on the Comprehensive R Archive Network (CRAN) https://CRAN.R-project.org/package=AQEval;
- The developers' version and code are publicly on GitHub https://github.com/karlropkins/AQEval, where issues or change requests can also be posted; and
- The project website is at https://karlropkins.github.io/AQEval/.

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Analytical Rationale

The main steps of the Break-Point/Segment (BP/S) analysis are:

1. Breaks-points are determined using the `strucchange` methods of Zeileis and colleagues (Zeileis et al., 2002, 2003). Here, a rolling-window approach is applied: a first subset (time-series window TW_0 in Figure 1a) is selected and linear regression modelled; the window advanced (TW_1 in Figure 1a) and a second model built, and so on through the time-series; then, likely points-of-change assigned by comparing the F-Stat scores of sequential models.
2. In addition to the standard Bayesian Information Criterion (BIC) testing used by `strucchange`, `AQEval` also checks all individual break-points are statistically valid ($p<0.05$), and down-scores less likely combinations.
3. Finally, the `segmented` methods of Muggeo and colleagues (Muggeo, 2003, 2017; Muggeo & others, 2008) are used to determine regions-of-change about break-points. Here, the confidence intervals for the selected break-points are used as start points, and segments assigned by random walk testing about these points as shown in Figure 1b.

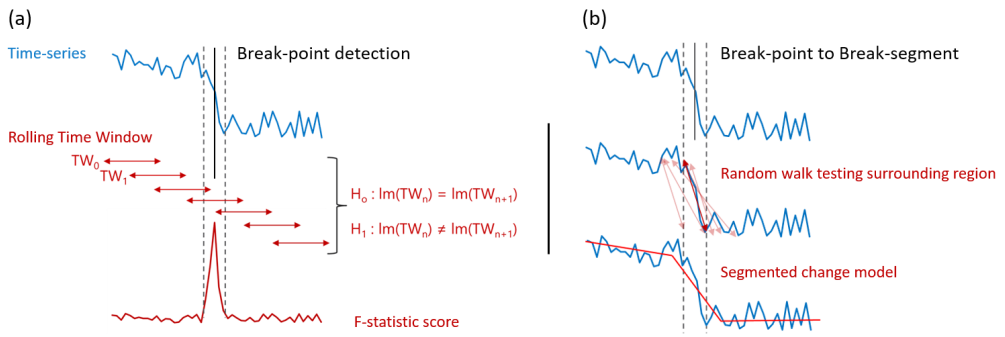

Figure 1: Break-point/segment scheme: (a) break-point, and (b) segment modelling about break-points.

Figure 2 shows the break-point/segment analysis of an NO_2 time-series from a roadside site where a change event (*ca.* 25 μg.m^{-3}; 31%) is detected between 2003-01-11 and 2003-02-19.

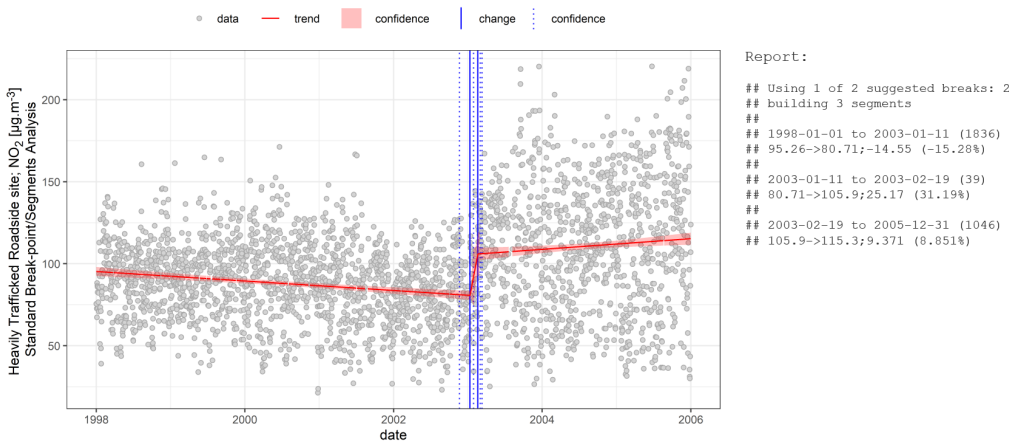

Figure 2: Standard AQEval break-point/segment analysis (graphical output and report) of an NO_2 1998-2005 time-series from a heavily trafficked roadside in the UK.

In some cases changes are small or local air inputs are complex, and time-series may require additional pre-processing to successfully isolate obscured change-events. For these, `AQEval` uses Generalized Additive Models (GAMs) (using `mgcv` methods, Wood, 2017, 2025) to subtract associated variance, by default:

$$[pollutant] = s_1(day\ of\ year) + s_2(hour\ of\ day) + te_1(wind\ speed, wind\ direction)$$

$$[pollutant]_{isolated} = ([pollutant] - [pollutant]_{predicted}) + mean(pollutant)$$

Where the investigated pollutant concentration, *[pollutant]*, is modelled as a function of *day of year*, *hour of day* and *wind speed* and *direction* using a combination of spline (s_1 and s_2) and tensor (te_1) fit-terms, and the unmodelled component, $[pollutant]_{isolated}$, is estimated as the mean-centred residual of this model.

Figure 3a shows the break-point analysis of NO_2 from a nearby but less heavily trafficked site where seasonality dominates the time-series, and Figure 3b shows the smaller (*ca.* 6.6 μg.m^{-3}; 13%) underlying change-event observed at a similar time to the large change observed at the more heavily trafficked site in Figure 2 using signal isolation and then break-point/segment analysis (2002-09-09 to 2002-12-21 compared with 2003-01-11 and 2003-02-19).

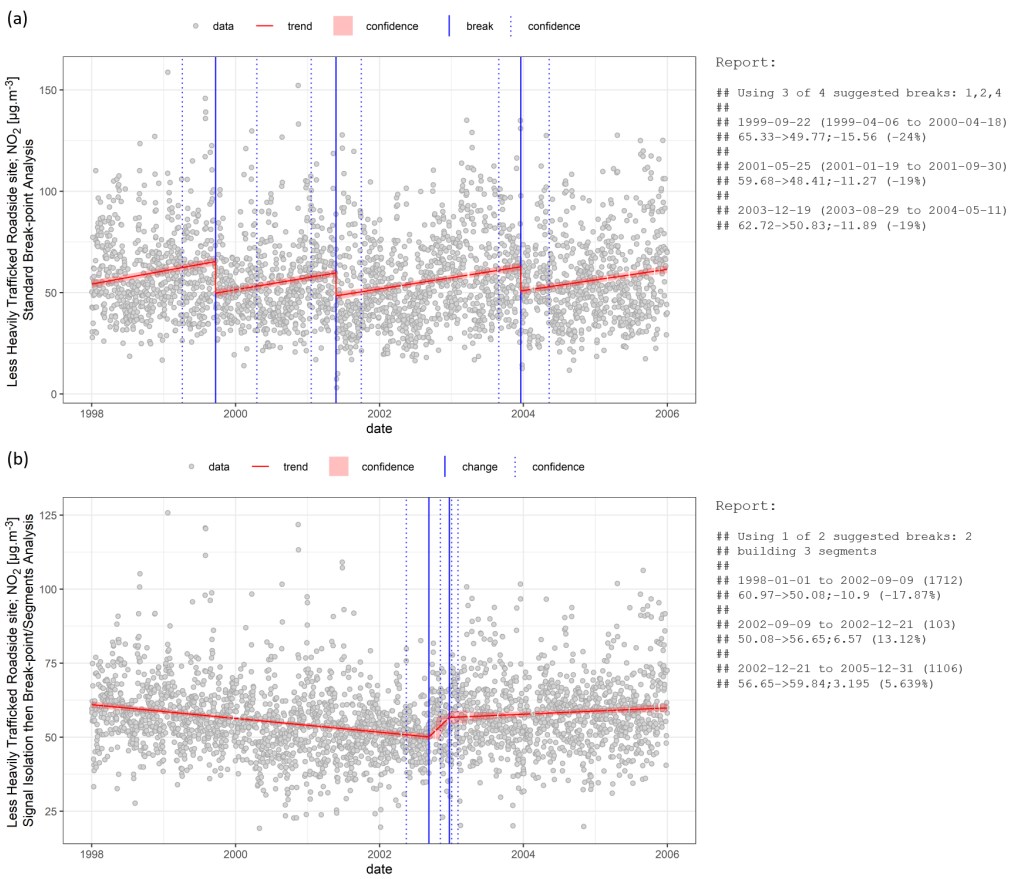

Figure 3: AQEval analysis of NO_2 1998-2005 time-series from a roadside site where: (a) standard (ambient air) break-point analysis exhibits a near-regular distribution of breaks typical of a site dominated by seasonal factors; and, (b) an underlying change-event is revealed using signal isolation and then break-point/segment analysis.

This default correction can also be modified to include other potential confounders, e.g. other frequency terms (e.g. day of week and/or week of year), background contributions (as local

variance associated with trends at near-by site not affected by the investigated change), or proxies for other local contributors (e.g. other meteorological parameters like air temperature, markers for other sources, etc).

Data Requirements and Main Functions

As many air quality professionals already use the `R` package `openair` (Carslaw & Ropkins, 2012; Ropkins & Carslaw, 2012) for more conventional analysis and data visualisation, `AQEval` uses `openair` coding syntax and data structures. Most importantly, time-series to be analysed should be provide `data.frame` (or similar), also containing a paired column of `POSIX*` class time-stamps named `date` similar to packaged example data, `aq.data`.

Main Functions:

```
# to quantify Break-Points (as in Figure 3a)
quantBreakPoints(
  data,               # data source, typically an openair-friendly data.frame
  pollutant,          # name of column containing the time-series to analysed
  h,                  # rolling window size, as proportion of time-series length
  ...                 # other arguments
)

# to quantify Break-Segments (as in Figure 2 and 3b)
quantBreakPoints(
  data,               # as above but fits trends to regions about break-points
  pollutant,          # (see section 3 in Analytical Rationale above)
  h,
  ...
)

# For signal isolation (as used for Figure 3b)
isolateContribution(
  data,
  pollutant,
  ...                 # other arguments allow user to modify the isolation method
)                     # (see package documentation for further details)

# findBreakPoints and testBreakPoints can also be used to fine-tune BP/S models
```

Related Outputs

The `AQEval` functions are described in further detail, along with worked examples of the code used to generate Figures 2 and 3, in the extended package introduction. Other work using `AQEval` include:

- Ropkins & Tate (2021), a peer-reviewed article on the multi-species AQEval analysis of air quality during the UK COVID-19 lockdown.
- Ropkins et al (2022), a peer-reviewed article on the use of `AQEval` to measure the NO_2 impact of bus fleet interventions.
- Also Clear Air Zone (CAZ) impact assessment reports include analyses using `AQEval`, see e.g.:
  - CAZ Baseline Study
  - First Year Report
  - (...)
  - Report archive

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